

MTH302 - Midsem 1

1. $(3 + 2)$ Compute the determinant of the matrix

$$\begin{pmatrix} 11 & 12 & 13 & 14 & 15 \\ 21 & 22 & 23 & 24 & 25 \\ 31 & 32 & 33 & 34 & 35 \\ 41 & 42 & 43 & 44 & 45 \\ 51 & 52 & 53 & 54 & 55 \end{pmatrix}$$

explaining each step in your computation.

Let $\mathbb{F}$ be a subfield of $\mathbb{C}$. Define n linear functionals on $\mathbb{F}^n$ ($n \geq 2$) by

$$f_k(x_1, \dots, x_n) = \sum_{j=1}^n (10k + j)x_j, \quad 1 \leq k \leq n$$

What is the dimension of the subspace annihilated by $f_1, \dots, f_n$?

2. $(2 + 2 + 2)$ Let V be the vector space of real polynomials of degree less than or equal to 2.
- (a) For $a \in \mathbb{R}$, let L_a denote evaluation at a . Prove that $B_L = \{L_1, L_2, L_3\}$ form a basis of V^* , the space of linear functionals on V .
 - (b) Determine a basis of V dual to B_L
 - (c) Let $f(x) \in V$ be such that

$$f(1) = 6, f(2) = 9, f(3) = 16$$

Determine the polynomial $f(x)$

3. (4 marks total) Give short explanation for your answers to the following:
- (a) (2 marks) Let V be the vector space of all real polynomials (viewed as functions). For real numbers $a < b$, define a linear functional $\gamma_{a,b}$ on V by $\gamma_{a,b}(f) = \int_a^b f(t)dt$. Let $D : V \rightarrow V$ denote differentiation. Compute $D^t \gamma_{a,b}$.
 - (b) (2 marks) Let $F = \mathbb{F}_5$. Find the number of 3×3 matrices over F of rank exactly 2.